

TS - 1.

a)  $h(n) = \delta(n) - \delta(n-4) \Rightarrow h(n) = \begin{cases} 1 & n=0 \\ -1 & n=4 \\ 0 & \text{ON OTRO CASO} \end{cases}$

$y(n) = x(n) * h(n)$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k)$$

$$= \sum_{k=0}^4 h(k) \cos(\omega_0 T_s (n-k))$$

$= \cos(\omega_0 T_s n) - \cos(\omega_0 T_s (n-4))$

$= \cos(\omega_0 T_s n) - \cos(\omega_0 T_s n) \cos(4\omega_0 T_s) - \sin(\omega_0 T_s n) \sin(4\omega_0 T_s)$

$= \underbrace{\cos(\omega_0 T_s n)}_{\alpha} [1 - \underbrace{\cos(4\omega_0 T_s)}_{\beta}] + \sin(\omega_0 T_s n) \sin(4\omega_0 T_s)$

$\alpha$

$\beta$

Según pide el enunciado.

$\Rightarrow \cos(\alpha) (1 - \cos(\beta)) + \sin(\alpha) \sin(\beta) = A \cos(\alpha + \phi)$

$\cos(\alpha) (1 - \cos(\beta)) + \sin(\alpha) \sin(\beta) = A \cos \alpha \cos \phi + A \sin \alpha \sin \phi$

$\begin{cases} A \cos \phi = 1 - \cos \beta \\ A \sin \phi = \sin \beta \end{cases}$

$A^2 \cos^2 \phi = (1 - \cos \beta)^2 \quad \text{y} \quad A^2 \sin^2 \phi = \sin^2 \beta$

$A^2 \cos^2 \phi + A^2 \sin^2 \phi = (1 - \cos \beta)^2 + \sin^2 \beta$

$A^2 (\cos^2 \phi + \sin^2 \phi) = (1 - \cos \beta)^2 + \sin^2 \beta$

$1$

$\Rightarrow A = \sqrt{(1 - \cos \beta)^2 + \sin^2 \beta} = \sqrt{2 - 2 \cos(4\omega_0 T_s)}$

$\phi = \tan^{-1} \frac{\sin(4\omega_0 T_s)}{1 - \cos(4\omega_0 T_s)}$

$$b) \quad x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$u(n) = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$$

$$y(n) = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^{n-k} u(n-k) \cdot h(k) =$$

$$= \left(\frac{1}{2}\right)^n \cdot u(n) - \left(\frac{1}{2}\right)^{n-4} \cdot u(n-4) = \left(\frac{1}{2}\right)^n \cdot u(n) - \left(\frac{1}{2}\right)^n \cdot 2^4 u(n-4)$$

$$= \left(\frac{1}{2}\right)^n (u(n) - 16 u(n-4))$$

$$c) \quad x(n) = u(n+1) - u(n-2)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x_k \cdot h(n-k) = \sum_{k=0}^4 h(k) \cdot x(n-k) =$$

$$= \sum_{k=0}^4 h(k) [u(n-k+1) - u(n-k-2)]$$

$$= (u(n+1) - u(n-2)) - (u(n-3) - u(n-6))$$